# STATISTICS AND PROBABILITY

ST331
(Statistical Computing 3)

# DISCRETE DISTRIBUTIONS

**Examples of discrete distributions are:**

- Bernoulli distribution
- Binomial distribution
- Poisson distribution

# BERNOULLI DISTRIBUTION

## Bernoulli distribution:

The Bernoulli distribution is the probability distribution of a discrete random variable that has only two possible outcomes, such as success or failure. This type of variable can be referred to as binary

$$f(x) = p^x(1-p)^{1-x}; \quad x = \{0,1\}$$

Say you use the common example of rolling a die, with success defined as getting a 4, and you roll once

```
> dbinom(x=1,size=1,prob=1/6)
[1] 0.1666667
```

# BINOMIAL DISTRIBUTION

- The binomial distribution is the distribution of successes in n number of trialsinvolving binary discrete random variables

$$f(x) = \binom{n}{x} p^x (1-p)^{n-x}; \quad x = \{0, 1, \ldots, n\}$$

where

$$\binom{n}{x} = \frac{n!}{x!(n-x)!}$$

- dbinom directly provides the mass function probabilities Pr(X = x) for any valid x—that is, 0 ≤ x ≤ n.
- pbinom provides the cumulative probability distribution—given a valid x,it yields Pr(X ≤ x).

# BINOMIAL DISTRIBUTION

I'll return to the example of rolling a die with success defined as getting a 4. If you roll the die independently eight times, what is the probability of observing exactly five successes (five 4s) in total?

```
> dbinom(x=5,size=8,prob=1/6)
[1] 0.004167619
```

```
> X.prob <- dbinom(x=0:8,size=8,prob=1/6)
> X.prob
[1] 2.325680e-01 3.721089e-01 2.604762e-01 1.041905e-01 2.604762e-02 4.167619e-03 4.167619e-04 2.381497e-05 5.953742e-07
> sum(X.prob)
[1] 1
> round(X.prob,3)
[1] 0.233 0.372 0.260 0.104 0.026 0.004 0.000 0.000 0.000
```

To find the mean and variance

```
> n<-8
> p<-1/6
> q<-5/6
> mu_x<-n*p
> mu_x
[1] 1.333333
> var_x<-n*p*q
> var_x
[1] 1.111111
```

# BINOMIAL DISTRIBUTION

To find, for example, the probability that youobserve three or fewer 4s, Pr(X ≤ 3

```
> sum(dbinom(x=0:3,size=8,prob=1/6))
[1] 0.9693436
> pbinom(q=3,size=8,prob=1/6)
[1] 0.9693436
```

To find the probability that you observe at least three 4s in eight rolls of the die, Pr(X ≥ 3)

```
> 1-pbinom(q=2,size=8,prob=1/6)
[1] 0.1348469
```

# POISSON DISTRIBUTION

$$f(x) = \frac{\lambda_p^x \exp(-\lambda_p)}{x!}; \quad x = \{0, 1, \ldots\}$$

A call center receives an average of 5 calls per hour.

Assume the number of calls follows a Poisson distribution.

(a) What is the probability that exactly 3 calls are received in one hour?

(b) What is the probability that more than 4 calls are received in one hour?

```
> 1-ppois(q = 4,lambda=5)
[1] 0.5595067
> dpois(x=3,lambda=5)
[1] 0.1403739
> 1-ppois(q = 4,lambda=5)
[1] 0.5595067
```

# CONTINUOUS DISTRIBUTIONS

We can characterise the continuous distributions by words as probability distribution whose support is an uncountable set such as intervals in the real line

Examples of continuous distribution are:

- Uniform distribution
- Exponential distribution
- Normal distribution

# UNIFORM DISTRIBUTION

$$f(x) = \begin{cases} \frac{1}{b-a} & \text{if } a \leq x \leq b; \\ 0 & \text{otherwise} \end{cases}$$

Customers arrive at a coffee shop at a random time between 9:00 AM and 9:20 AM. Assume the arrival time is uniformly distributed. Let X = arrival time in minutes after 9:00 AM. X~Uniform(0,20)

What is the probability that a customer arrives within the first 8 minutes?

What is the probability that a customer arrives after 15 minutes?

```
> punif(8, min = 0, max = 20)
[1] 0.4
>
> 1 - punif(15, min = 0, max = 20)
[1] 0.25
```

# EXPONENTIAL DISTRIBUTION

$$f(x) = \begin{cases} \lambda e^{-\lambda x} & x \geq 0 \\ 0 & x < 0 \end{cases}$$

$$\mu_X = \frac{1}{\lambda_e} \quad \text{and} \quad \sigma_X^2 = \frac{1}{\lambda_e^2}$$

The time (in minutes) between arrivals of customers at a bank follows an exponential distribution with an average of 5 minutes between arrivals. Let X be the time until the next customer arrives.

What is the probability that the next customer arrives within 3 minutes?
What is the probability that the waiting time exceeds 7 minutes?

```
> pexp(q=3, rate = 0.2)
[1] 0.4511884
> 1 - pexp(q=7, rate = 0.2)
[1] 0.246597
```

# NORMAL DISTRIBUTION

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}.$$

Assume that the test scores of a college entrance exam follow a normal distribution. Furthermore, the mean test score is 70 and the standard deviation is 10. What percentage of students:

a) score 85 or more on the exam?

b) score 60 or less on the exam?

```
> pnorm (85, mean =70, sd=10, lower.tail=FALSE)
[1] 0.0668072
> pnorm (60, mean =70, sd =10)
[1] 0.1586553
> pnorm (60, mean =70, sd =10, lower.tail=TRUE)
[1] 0.1586553
```

# NORMAL DISTRIBUTION

The heights of students in a university are normally distributed with a mean of 170 cm and a standard deviation of 8 cm. What percentage of students have heights between 165 cm and 182 cm?

```
> pnorm(182, mean = 170, sd = 8) - pnorm(165, mean = 170, sd = 8)
[1] 0.6672073
```

# CLASS EXERCISE

ST431

(Design and Analysis of Experiment)